CSA0921 - Programming in Java
Level 3 Practice Programs

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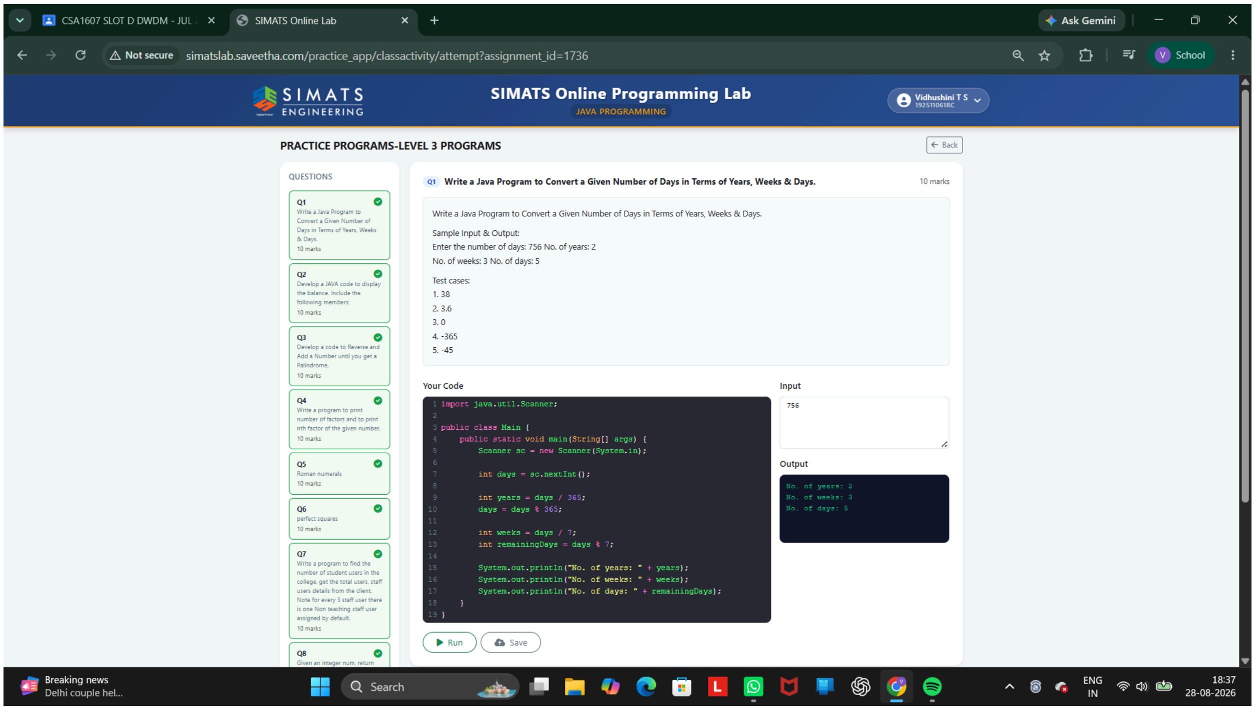
1.



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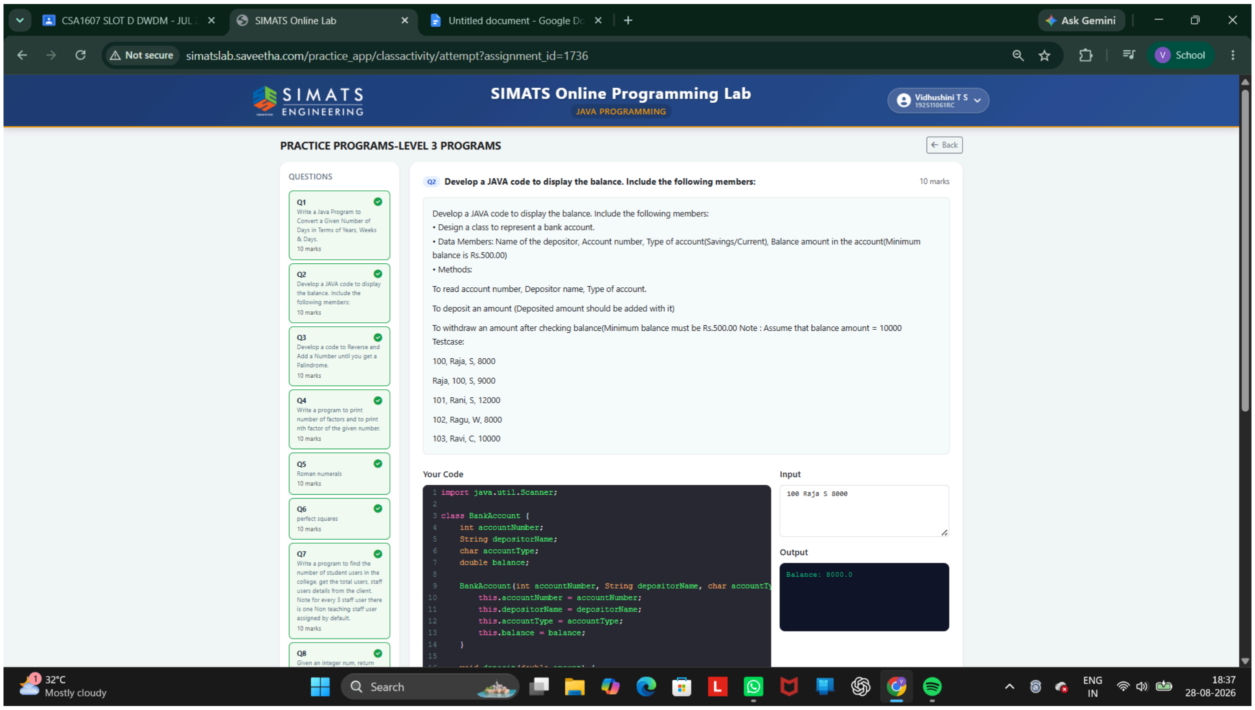
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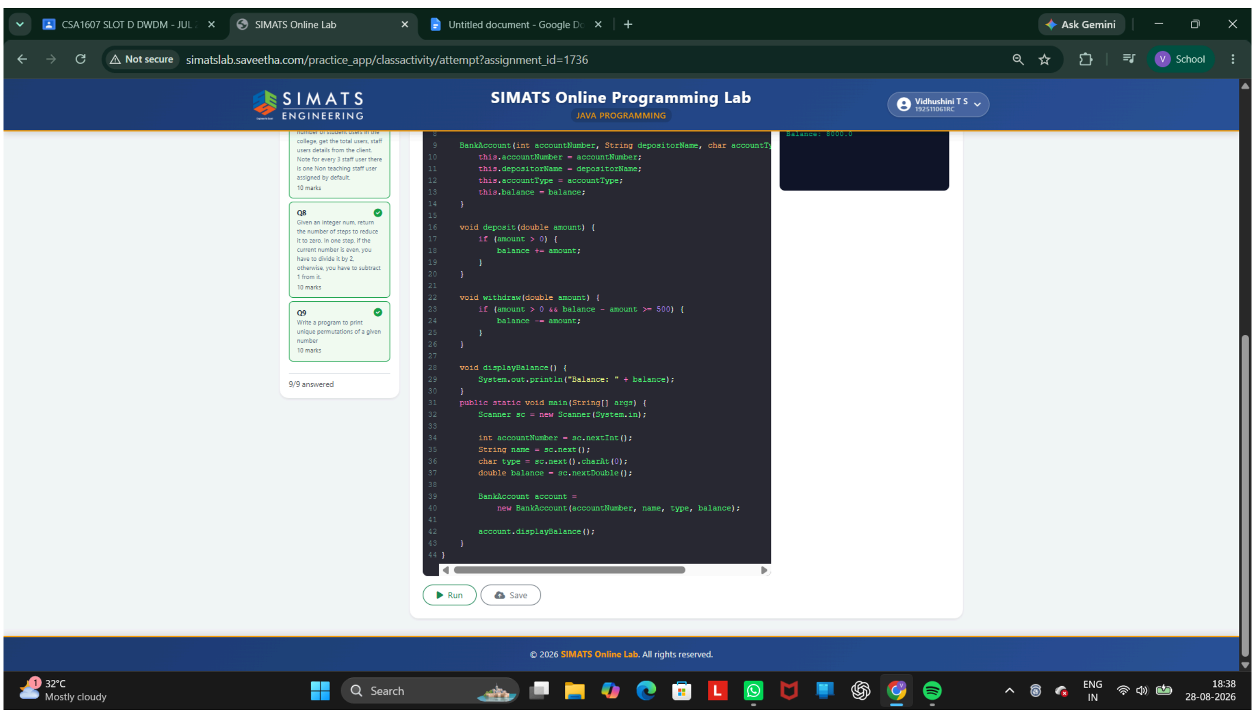
PRACTICE PROGRAMS-LEVEL 3 PROGRAMS
Q2 Develop a JAVA code to display the balance. Include the following members:
10 marks
Develop a JAVA code to display the balance. Include the following members:
• Design a class to represent a bank account.
• Data Members: Name of the depositor, Account number, Type of account(Savings/Current), Balance amount in the account(Minimum balance is Rs.500.00)
• Methods:
To read account number, Depositor name, Type of account.
To deposit an amount (Deposited amount should be added with it)
To withdraw an amount after checking balance(Minimum balance must be Rs.500.00 Note : Assume that balance amount = 10000
Testcase:
100, Raja, S, 8000
Raja, 100, S, 9000
101, Rani, S, 12000
102, Ragu, W, 8000
103, Ravi, C, 10000
Your Code
import java.util.Scanner;
class BankAccount {
int accountNumber;
String depositorName;
char accountType;
double balance;
BankAccount(int accountNumber, String depositorName, char accountTy
this.accountNumber = accountNumber;
this.depositorName = depositorName;
this.accountType = accountType;
this.balance = balance;
}
Input
100 Raja S 8000
Output
Balance: 8000.0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BankAccount(int accountNumber, String depositorName, char accountTy
this.accountNumber = accountNumber;
this.depositorName = depositorName;
this.accountType = accountType;
this.balance = balance;
}
void deposit(double amount) {
if (amount > 0) {
balance += amount;
}
}
void withdraw(double amount) {
if (amount > 0 && balance - amount >= 500) {
balance -= amount;
}
}
void displayBalance() {
System.out.println("Balance: " + balance);
}
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int accountNumber = sc.nextInt();
String name = sc.next();
char type = sc.next().charAt(0);
double balance = sc.nextDouble();
BankAccount account =
new BankAccount(accountNumber, name, type, balance);
account.displayBalance();
}
}
Run
Save
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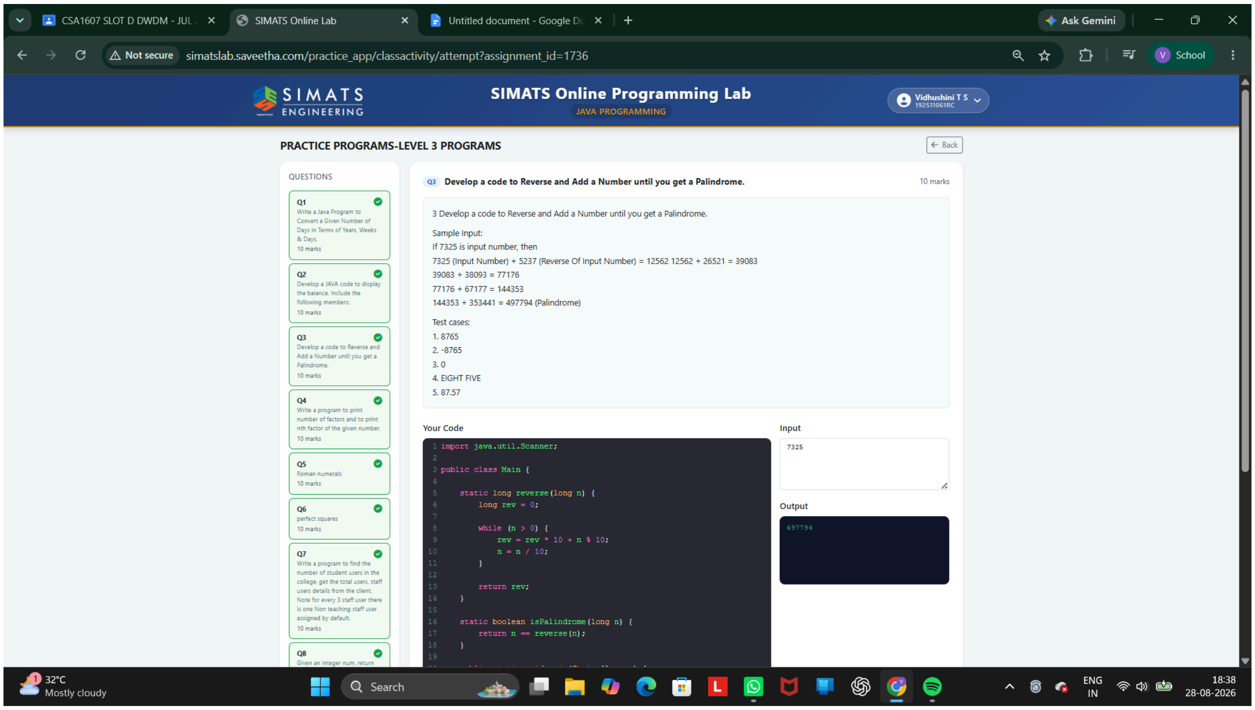
3.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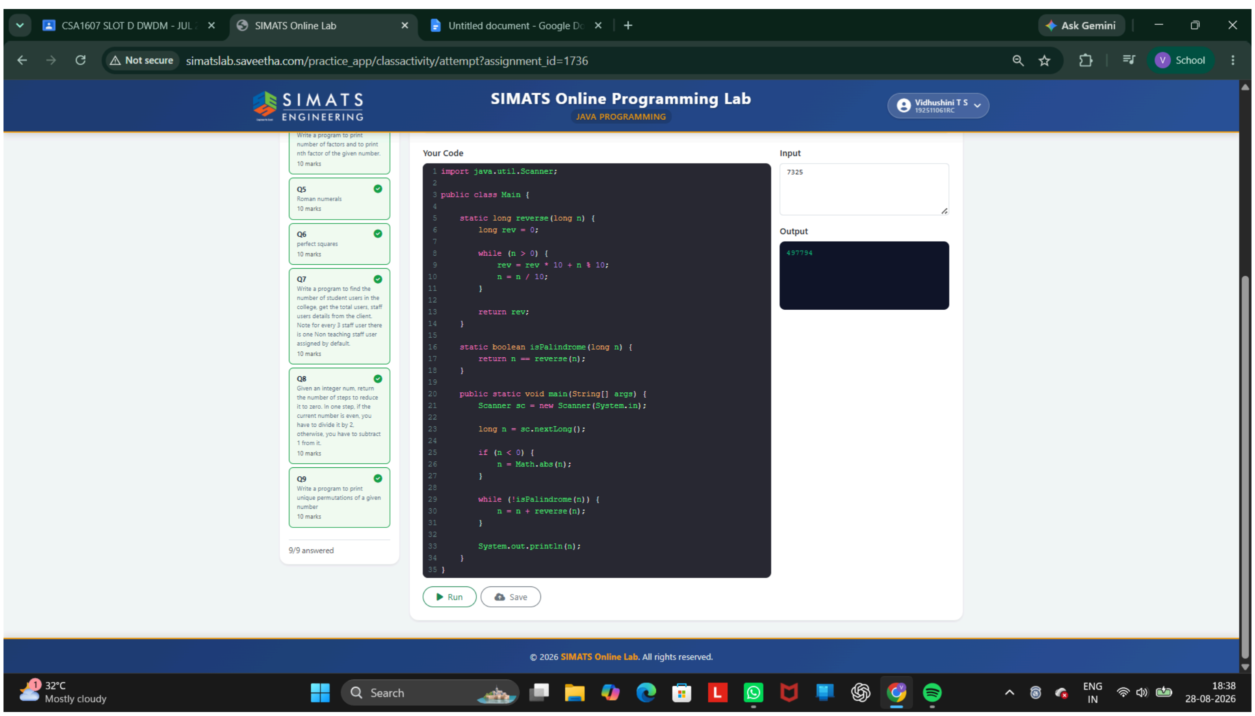
Q3 Develop a code to Reverse and Add a Number until you get a Palindrome.
10 marks
3 Develop a code to Reverse and Add a Number until you get a Palindrome.
Sample Input:
If 7325 is input number, then
7325 (Input Number) + 5237 (Reverse Of Input Number) = 12562 12562 + 26521 = 39083
39083 + 38093 = 77176
77176 + 67177 = 144353
144353 + 353441 = 497794 (Palindrome)
Test cases:
1. 8765
2. -8765
3. 0
4. EIGHT FIVE
5. 87.57
Your Code
Input
7325
Output
497794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Your Code
import java.util.Scanner;
public class Main {
static long reverse(long n) {
long rev = 0;
while (n > 0) {
rev = rev * 10 + n % 10;
n = n / 10;
}
return rev;
}
static boolean isPalindrome(long n) {
return n == reverse(n);
}
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
long n = sc.nextLong();
if (n < 0) {
n = Math.abs(n);
}
while (!isPalindrome(n)) {
n = n + reverse(n);
}
System.out.println(n);
}
}
Run
Save
Input
7325
Output
497794
9/9 answered
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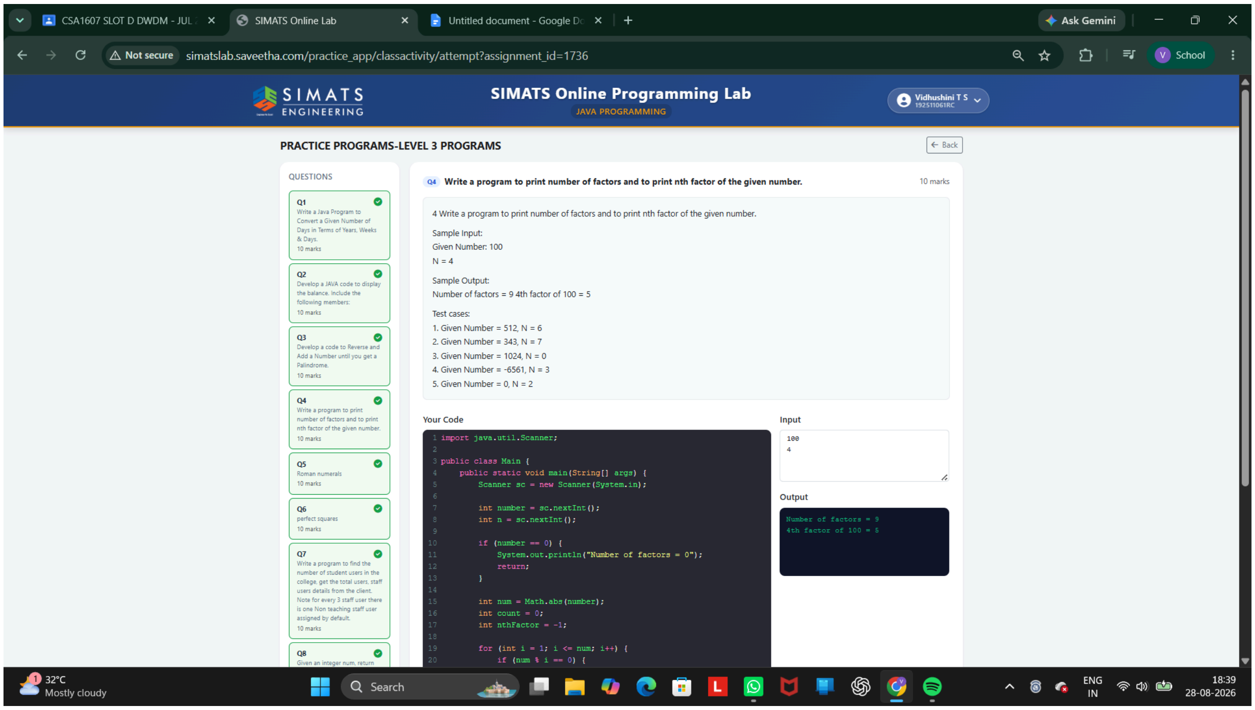
4.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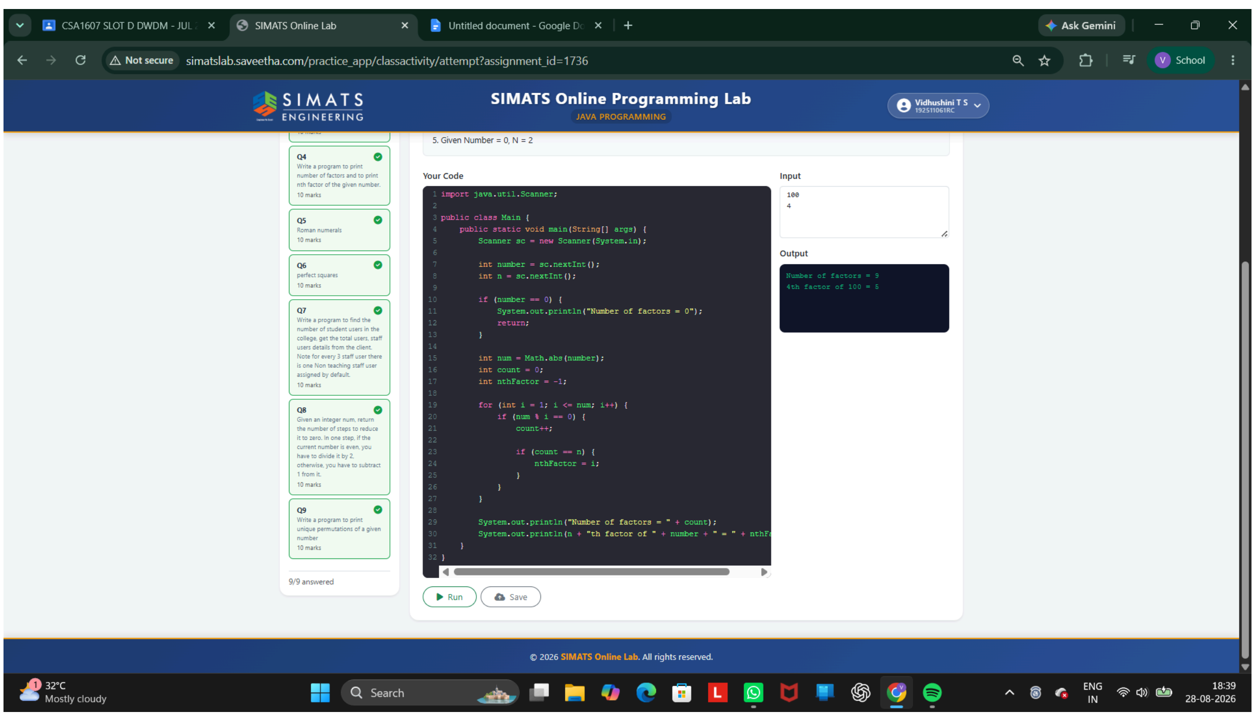
Q4 Write a program to print number of factors and to print nth factor of the given number.
10 marks
4 Write a program to print number of factors and to print nth factor of the given number.
Sample Input:
Given Number: 100
N = 4
Sample Output:
Number of factors = 9 4th factor of 100 = 5
Test cases:
1. Given Number = 512, N = 6
2. Given Number = 343, N = 7
3. Given Number = 1024, N = 0
4. Given Number = -6561, N = 3
5. Given Number = 0, N = 2
Your Code
Input
100
4
Output
Number of factors = 9
4th factor of 100 = 5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5. Given Number = 0, N = 2
Your Code
import java.util.Scanner;
public class Main {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int number = sc.nextInt();
int n = sc.nextInt();
if (number == 0) {
System.out.println("Number of factors = 0");
return;
}
int num = Math.abs(number);
int count = 0;
int nthFactor = -1;
for (int i = 1; i <= num; i++) {
if (num % i == 0) {
count++;
if (count == n) {
nthFactor = i;
}
}
}
System.out.println("Number of factors = " + count);
System.out.println(n + "th factor of " + number + " = " + nthFa
}
}
Input
100
4
Output
Number of factors = 9
4th factor of 100 = 5
Run
Save
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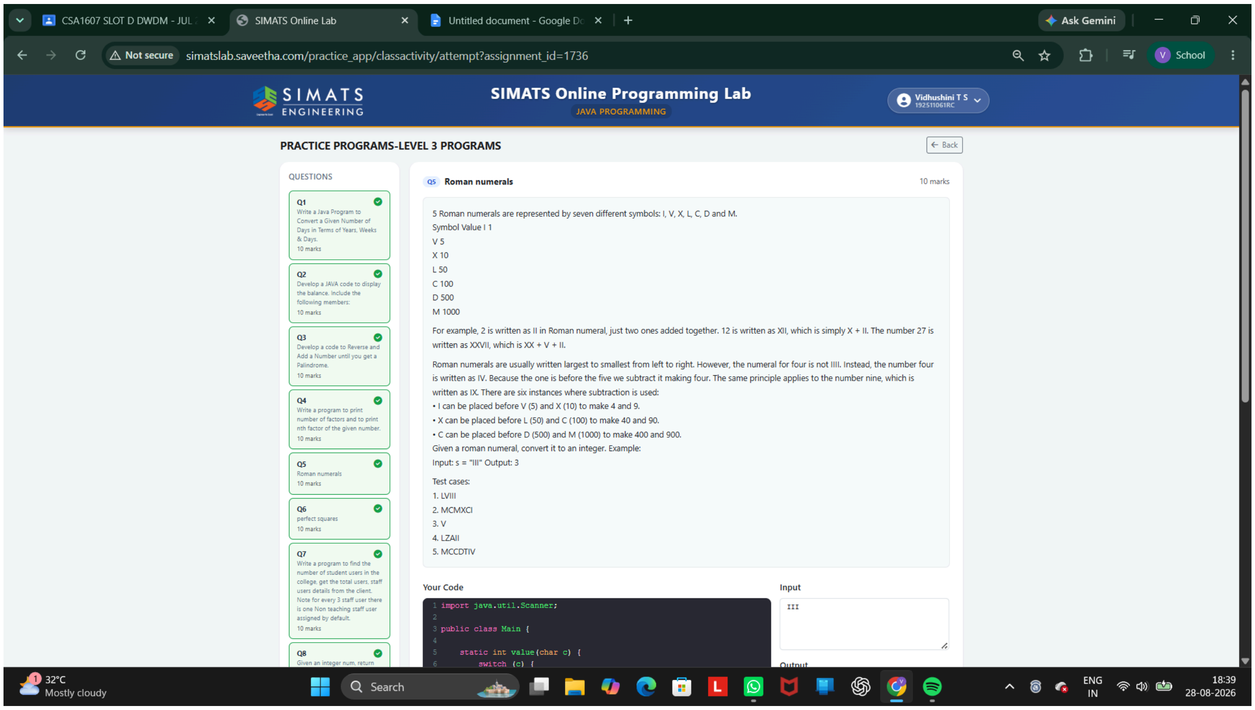
5.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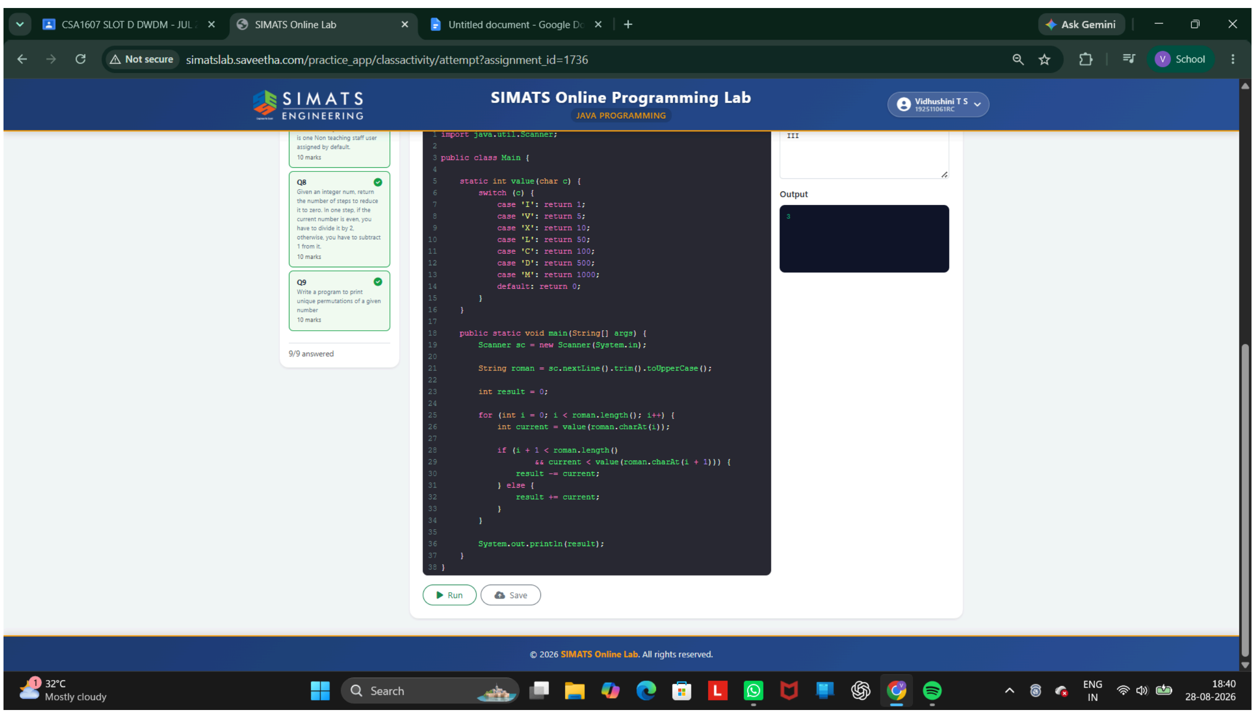
Q5 Roman numerals
10 marks
5 Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.
Symbol Value I 1
V 5
X 10
L 50
C 100
D 500
M 1000
For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is simply X + II. The number 27 is written as XXVII, which is XX + V + II.
Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:
• I can be placed before V (5) and X (10) to make 4 and 9.
• X can be placed before L (50) and C (100) to make 40 and 90.
• C can be placed before D (500) and M (1000) to make 400 and 900.
Given a roman numeral, convert it to an integer. Example:
Input: s = "III" Output: 3
Test cases:
1. LVIII
2. MCMXCI
3. V
4. LZAII
5. MCCDTIV
Your Code
Input
III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import java.util.Scanner;
public class Main {
static int value(char c) {
switch (c) {
case 'I': return 1;
case 'V': return 5;
case 'X': return 10;
case 'L': return 50;
case 'C': return 100;
case 'D': return 500;
case 'M': return 1000;
default: return 0;
}
}
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
String roman = sc.nextLine().trim().toUpperCase();
int result = 0;
for (int i = 0; i < roman.length(); i++) {
int current = value(roman.charAt(i));
if (i + 1 < roman.length()
&& current < value(roman.charAt(i + 1))) {
result -= current;
} else {
result += current;
}
}
System.out.println(result);
}
}
Output
3
Run
Save
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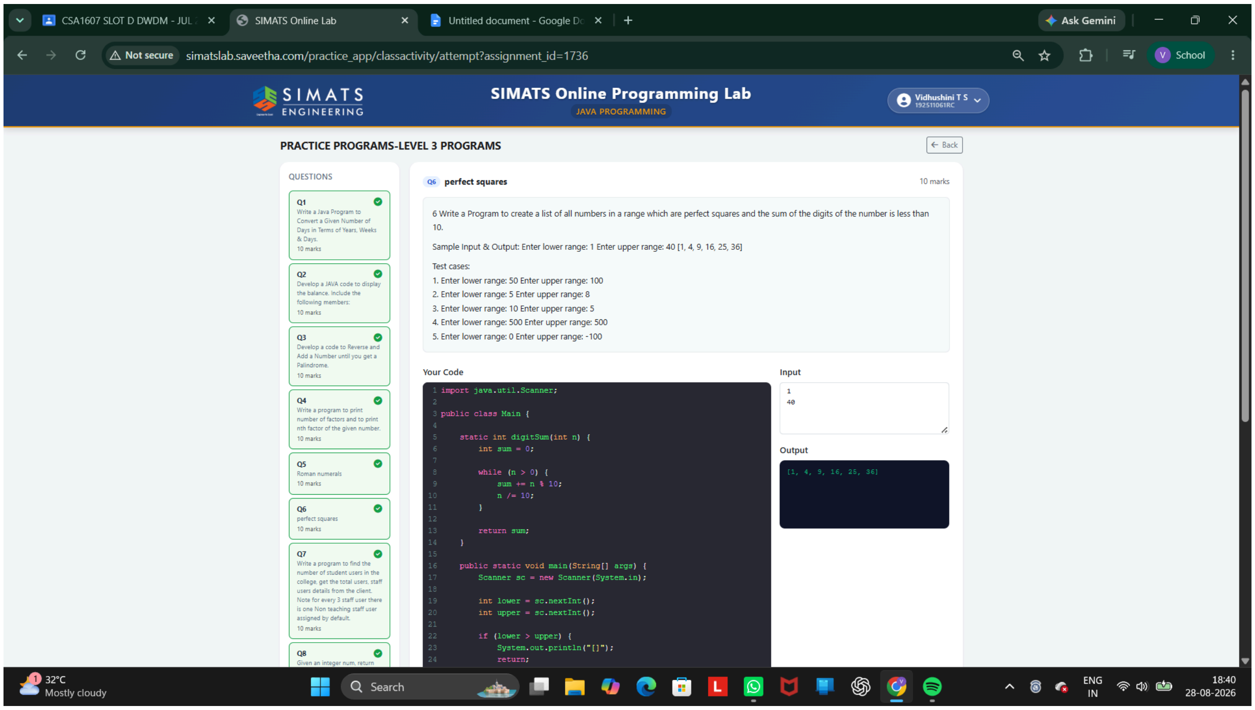
6.



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Q6 perfect squares

10 marks

6 Write a Program to create a list of all numbers in a range which are perfect squares and the sum of the digits of the number is less than 10.

Sample Input & Output: Enter lower range: 1 Enter upper range: 40 [1, 4, 9, 16, 25, 36]

Test cases:

1. Enter lower range: 50 Enter upper range: 100
2. Enter lower range: 5 Enter upper range: 8
3. Enter lower range: 10 Enter upper range: 5
4. Enter lower range: 500 Enter upper range: 500
5. Enter lower range: 0 Enter upper range: -100

Your Code

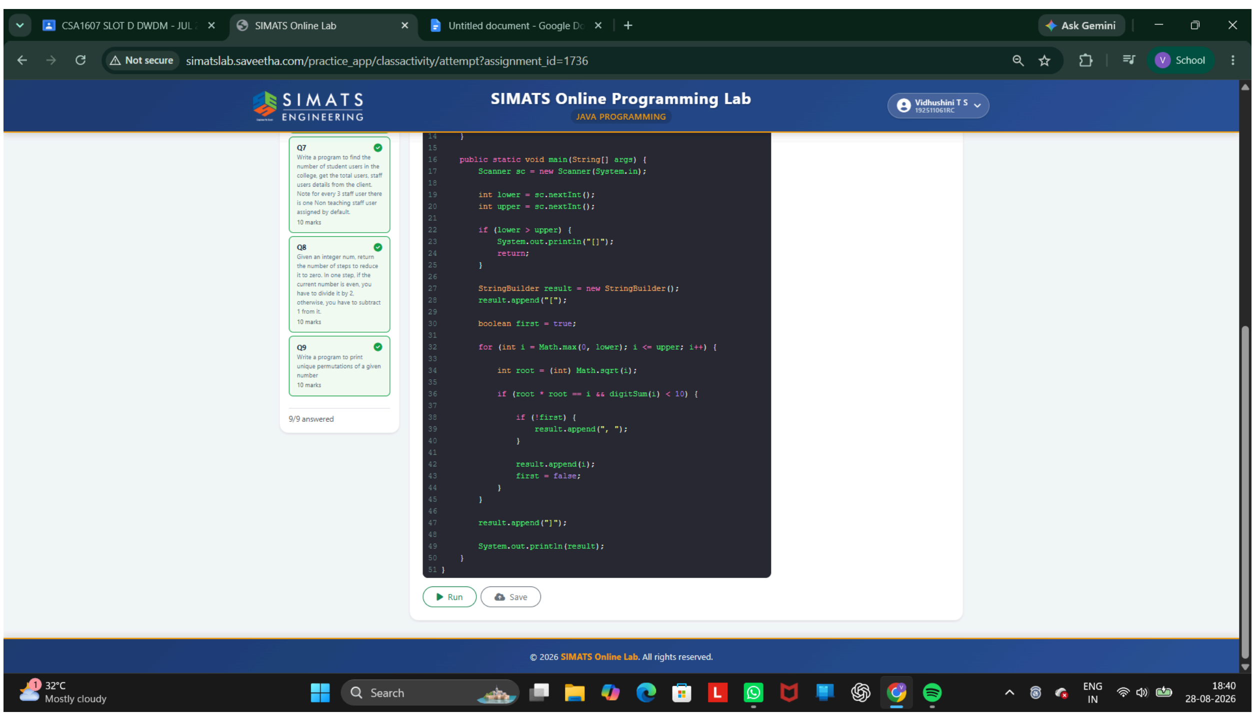
```java
import java.util.Scanner;

public class Main {

    static int digitSum(int n) {
        int sum = 0;

        while (n > 0) {
            sum += n % 10;
            n /= 10;
        }

        return sum;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int lower = sc.nextInt();
        int upper = sc.nextInt();

        if (lower > upper) {
            System.out.println("[]");
            return;
```

Input

```
1
40
```

Output

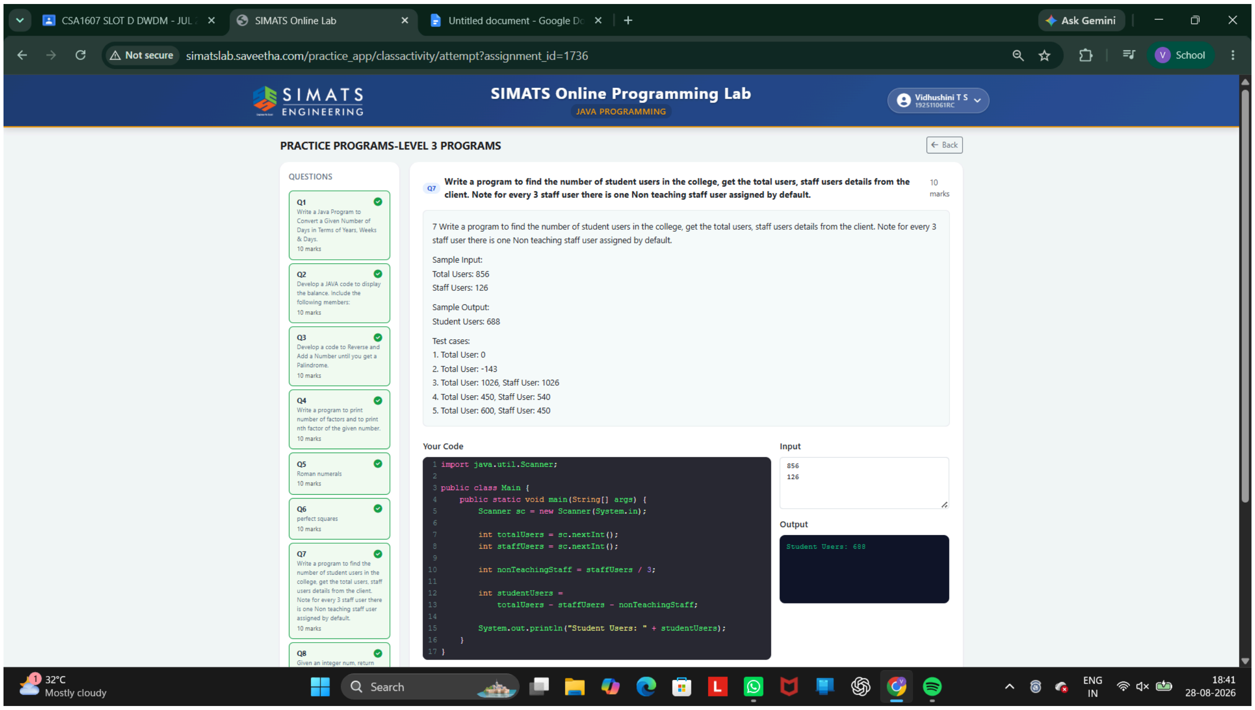
```
[1, 4, 9, 16, 25, 36]
```



```java
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int lower = sc.nextInt();
        int upper = sc.nextInt();

        if (lower > upper) {
            System.out.println("[]");
            return;
        }

        StringBuilder result = new StringBuilder();
        result.append("[");

        boolean first = true;

        for (int i = Math.max(0, lower); i <= upper; i++) {

            int root = (int) Math.sqrt(i);

            if (root * root == i && digitSum(i) < 10) {

                if (!first) {
                    result.append(", ");
                }

                result.append(i);
                first = false;
            }
        }

        result.append("]");

        System.out.println(result);
    }
}
```

Run Save

7.



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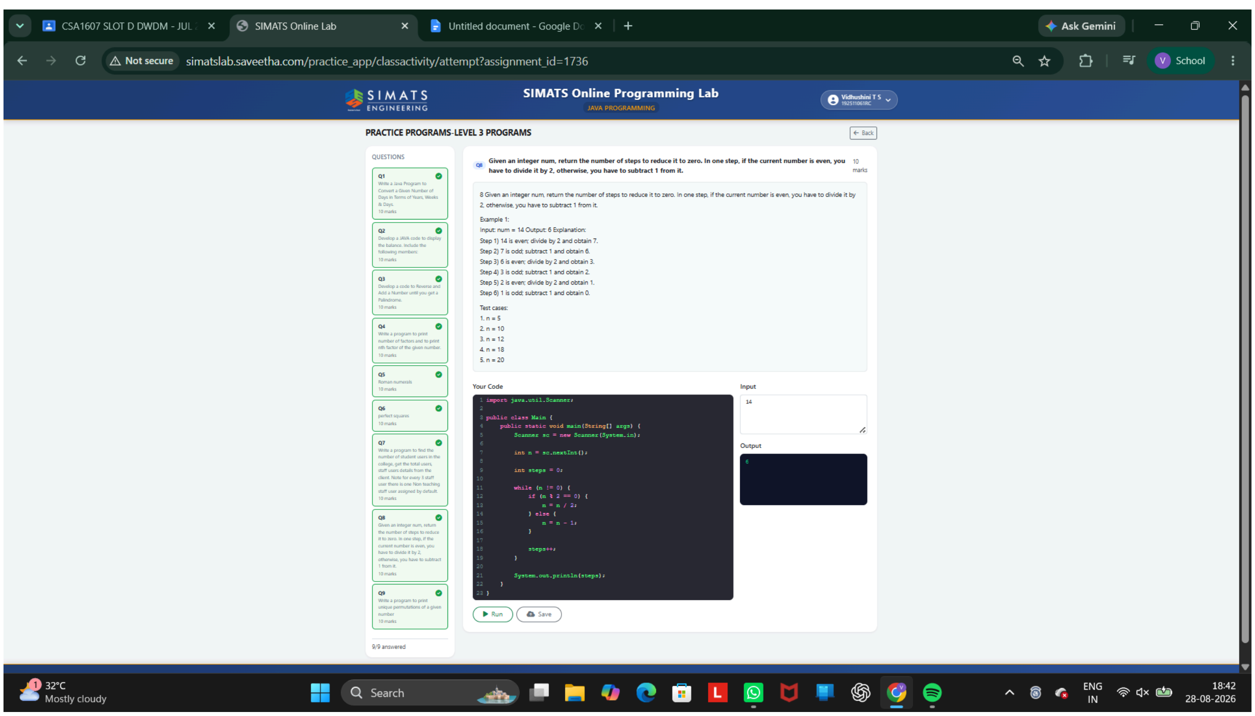
Back
QUESTIONS
Q7
Write a program to find the number of student users in the college, get the total users, staff users details from the client. Note for every 3 staff user there is one Non teaching staff user assigned by default.
10 marks
7 Write a program to find the number of student users in the college, get the total users, staff users details from the client. Note for every 3 staff user there is one Non teaching staff user assigned by default.
Sample Input:
Total Users: 856
Staff Users: 126
Sample Output:
Student Users: 688
Test cases:
1. Total User: 0
2. Total User: -143
3. Total User: 1026, Staff User: 1026
4. Total User: 450, Staff User: 540
5. Total User: 600, Staff User: 450
Your Code
import java.util.Scanner;
public class Main {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int totalUsers = sc.nextInt();
int staffUsers = sc.nextInt();
int nonTeachingStaff = staffUsers / 3;
int studentUsers =
totalUsers - staffUsers - nonTeachingStaff;
System.out.println("Student Users: " + studentUsers);
}
}
Input
856
126
Output
Student Users: 688

8.



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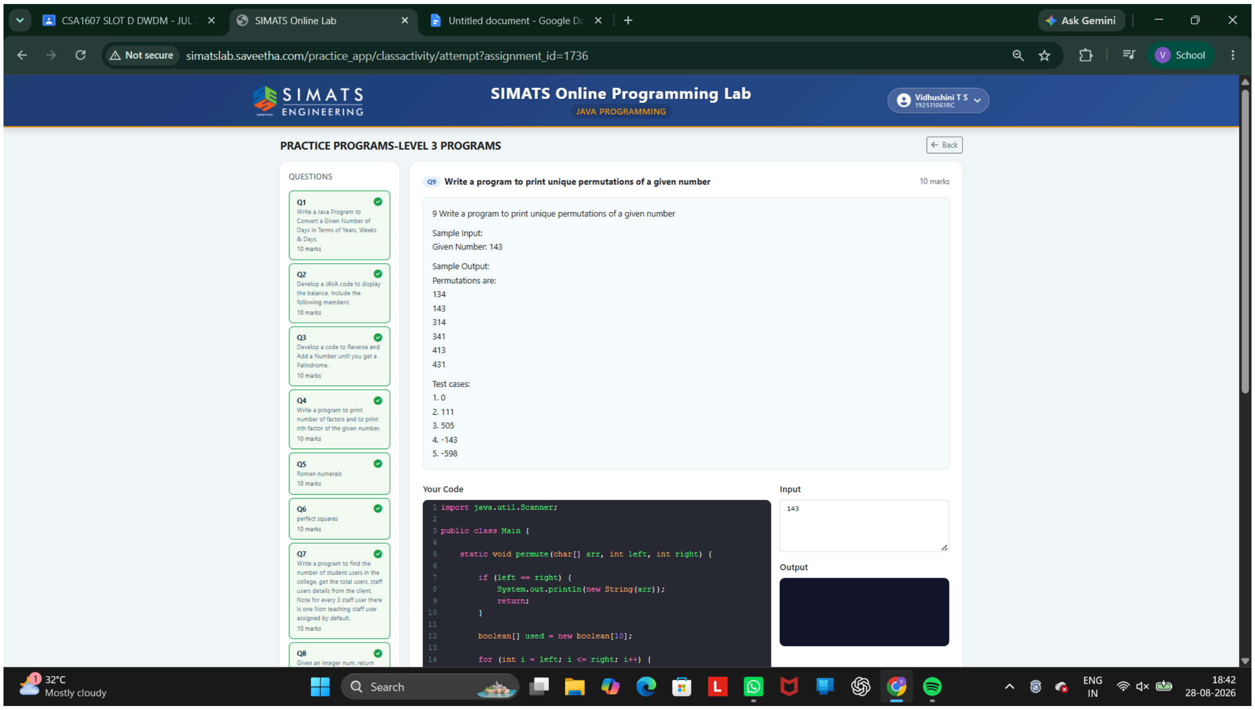
QUESTIONS
Q8
Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.
10 marks
8 Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.
Example 1:
Input: num = 14 Output: 6 Explanation:
Step 1) 14 is even; divide by 2 and obtain 7.
Step 2) 7 is odd; subtract 1 and obtain 6.
Step 3) 6 is even; divide by 2 and obtain 3.
Step 4) 3 is odd; subtract 1 and obtain 2.
Step 5) 2 is even; divide by 2 and obtain 1.
Step 6) 1 is odd; subtract 1 and obtain 0.
Test cases:
1. n = 5
2. n = 10
3. n = 12
4. n = 18
5. n = 20
Your Code
import java.util.Scanner;
public class Main {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int steps = 0;
while (n != 0) {
if (n % 2 == 0) {
n = n / 2;
} else {
n = n - 1;
}
steps++;
}
System.out.println(steps);
}
}
Run
Save
Input
14
Output
6
9/9 answered

9.



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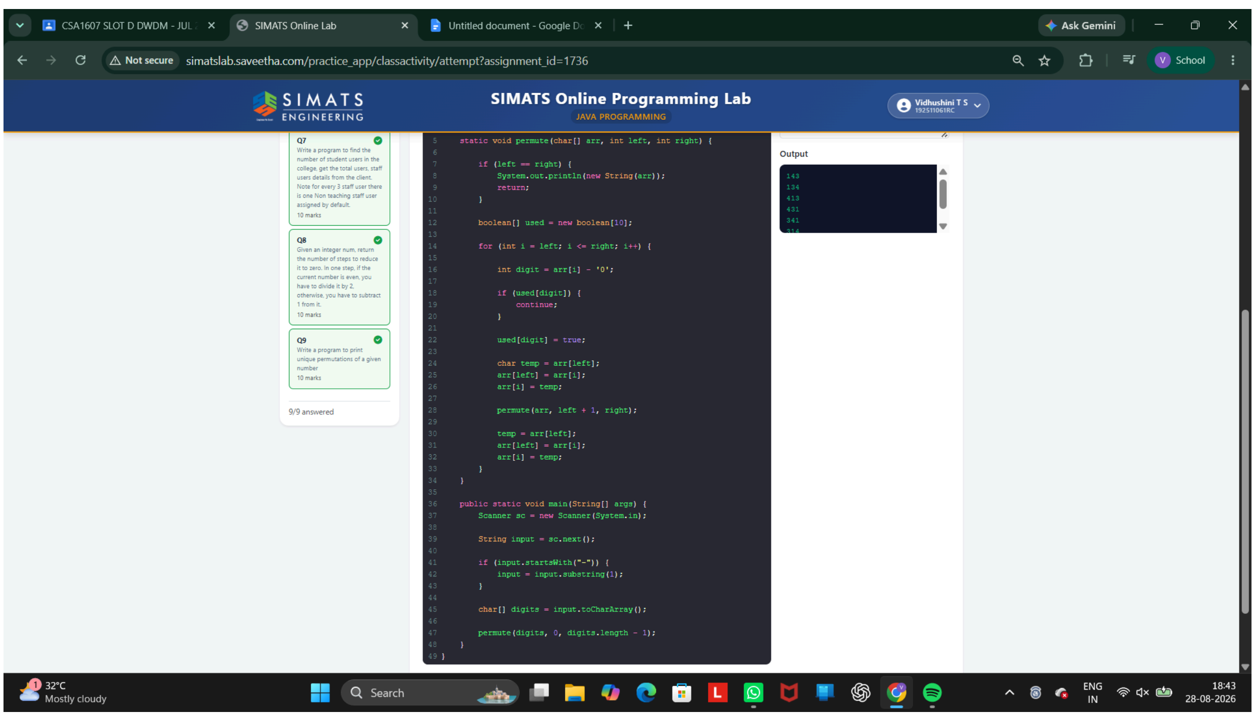
Back
QUESTIONS
Q1
Write a Java Program to Convert a Given Number of Days in Terms of Years, Weeks & Days.
10 marks
Q2
Develop a JAVA code to display the balance. Include the following members:
10 marks
Q3
Develop a code to Reverse and Add a Number until you get a Palindrome.
10 marks
Q4
Write a program to print number of factors and to print nth factor of the given number.
10 marks
Q5
Roman numerals
10 marks
Q6
perfect squares
10 marks
Q7
Write a program to find the number of student users in the college, get the total users, staff users details from the client. Note for every 3 staff user there is one Non teaching staff user assigned by default.
10 marks
Q8
Given an integer num, return
Q9
Write a program to print unique permutations of a given number
10 marks
9 Write a program to print unique permutations of a given number
Sample Input:
Given Number: 143
Sample Output:
Permutations are:
134
143
314
341
413
431
Test cases:
1. 0
2. 111
3. 505
4. -143
5. -598
Your Code
1 import java.util.Scanner;
2
3 public class Main {
4
5 static void permute(char[] arr, int left, int right) {
6
7 if (left == right) {
8 System.out.println(new String(arr));
9 return;
10 }
11
12 boolean[] used = new boolean[10];
13
14 for (int i = left; i <= right; i++) {
Input
143
Output
32°C
Mostly cloudy
Search
ENG
IN
18:42
28-08-2026



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Q7
Write a program to find the number of student users in the college, get the total users, staff users details from the client. Note for every 3 staff user there is one Non teaching staff user assigned by default.
10 marks
Q8
Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.
10 marks
Q9
Write a program to print unique permutations of a given number
10 marks
9/9 answered
5 static void permute(char[] arr, int left, int right) {
6
7 if (left == right) {
8 System.out.println(new String(arr));
9 return;
10 }
11
12 boolean[] used = new boolean[10];
13
14 for (int i = left; i <= right; i++) {
15
16 int digit = arr[i] - '0';
17
18 if (used[digit]) {
19 continue;
20 }
21
22 used[digit] = true;
23
24 char temp = arr[left];
25 arr[left] = arr[i];
26 arr[i] = temp;
27
28 permute(arr, left + 1, right);
29
30 temp = arr[left];
31 arr[left] = arr[i];
32 arr[i] = temp;
33 }
34 }
35
36 public static void main(String[] args) {
37 Scanner sc = new Scanner(System.in);
38
39 String input = sc.next();
40
41 if (input.startsWith("-")) {
42 input = input.substring(1);
43 }
44
45 char[] digits = input.toCharArray();
46
47 permute(digits, 0, digits.length - 1);
48 }
49 }
Output
143
134
413
431
341
32°C
Mostly cloudy
Search
ENG
IN
18:43
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